

C02-ATI

Analytical problem solving
ITAD502- Computer vision.

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① Grey level Transformation:

Given:

$$S = 255 - r$$

Pixel values: $r = 50, 100, 150, 200$

② purpose:

Grey level Transformation is used to improve the appearance of an image by changing the intensity of pixel values. It helps in enhancing brightness, contrast and image details.

③ Negative Transformation:

$$S = 255 - r$$

original pixel	Calculation	Transformed pixel
50	$255 - 50$	205
100	$255 - 100$	155
150	$255 - 150$	105
200	$255 - 200$	55

with the Importance of Edge detection, noise removal and

Transformed values: 205, 155, 105, 55

Interpretation:

- 1) Brighter pixel become darker
- * darker pixels become brighter.

2) Histogram equalization:

(a) Purpose:

Histogram equalization improves the contrast of an image by spreading pixel intensity values over a wider range.

Intensity	frequency
0	4
1	6
2	10
3	6

$$\text{Total pixels} = N = 4 + 6 + 10 + 6 = 26$$

(b) Cumulative distribution function (CDF)

Intensity	frequency	CDF
0	4	4
1	6	10
2	10	20
3	6	26

Equalized Intensity:

$$S = \frac{cdf}{N} \times (L-1)$$

$$N = 26$$

$$L = 4$$

$$S = \frac{cdf}{26} \times 3$$

Intensity	DF	Calculation	Equalized value
0	4	$(4/26) \times 3 = 0.46$	0
1	10	$(10/26) \times 3 = 1.15$	1
2	20	$(20/26) \times 3 = 2.31$	2
3	26	$(26/26) \times 3 = 3$	3

Interpretation:

* Intensity are distributed frequently.

3) Gaussian Smoothing:

purpose:

Gaussian filtering removes image noise while preserving edges. Better the average processing, it gives more weight to the central pixel.

$$(b) \text{ Kernel} = \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix} \quad \text{Image region} = \begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 40 & 20 & 10 \end{bmatrix}$$

$$\Rightarrow (10 \times 1) + (20 \times 2) + (10 \times 1) + (20 \times 2) + (40 \times 1) + (20 \times 2) + (10 \times 1) + (20 \times 2) + (10 \times 1)$$

$$\Rightarrow 10 + 40 + 10 + 40 + 10 + 20 + 10 + 20 + 10$$

$$\Rightarrow \frac{360}{16} = 22.5 \approx 23$$

$$\text{New center pixel} = 23$$

(d) Edge detection using Sobel operator:

(a) purpose:

Edge detection identifies sudden intensity change in an

image. It helps detect object boundaries and shapes.

(b) Horizontal Sobel mask:

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Image:

$$\begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

Top row:

$$(-1 \times 10) + (-2 \times 10) + (-1 \times 10) = -40$$

Middle row:

0

Bottom:

$$10 \times 1 \times 30 + 2 \times 30 + 1 \times 30 = 150$$

$$G = -40 + 150 = 80$$

$$\text{Gradient} = 80$$

⑤ Contrast Stretching:

(a) purpose:

Contrast stretching increases image contrast by expanding a narrow intensity range to the full grayscale range.

$$(b) \quad S = \frac{(r - r_{min})}{(r_{max} - r_{min})} \times 255$$

$$r_{min} = 50$$

$$r_{max} = 150$$

For $r = 80$

$$S = \frac{80 - 50}{150 - 50} \times 255$$

$\Rightarrow 0$

For $r = 100$

$$S = \frac{100 - 50}{100} \times 255$$

$$\Rightarrow 0.5 \times 255$$

$$\Rightarrow 127.5 \approx 128$$

For $r = 150$

$$S = \frac{150 - 50}{100} \times 255$$

$$\Rightarrow 255$$

original (r)	Transformed (s)
50	0
100	128
150	255